ROBERT SCHLEGEL, PhD

Environmental Data Scientist

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Toulouse, France



PROFILE

Results-oriented professional with over a decade of experience on international projects at the nexus of climate and ecology. As a resourceful problem solver, I draw upon my robust academic research background to develop practical solutions based on science and technology to support resilience against the adverse effects of climate change on both human communities and the environment.

CORE SKILLS

- R + Python programming languages
- Shiny, Quarto, HTML, SQL
- Full-stack app development
- Data acquisition pipelines
- Spatial and temporal analyses of big data
- Management, development, and publication of datasets
- Machine learning algorithm development
- Project management
- Curriculum development
- Teaching - Instructing - Demonstrating
- Mentoring
- Certified PADI diving instructor

CORE KNOWLEDGE

- Marine heatwaves (MHW)
- Coastal ecology - kelp forests
- Species distribution modelling (SDM)
- Climate change
- Biodiversity
- Languages: English, French, German

EXPERIENCE

Oceanographer – Validation Protocols | Mercator Ocean International

Toulouse - FR - Oct 2026 – Present

- Develop and document validation protocols for Copernicus Marine ocean forecasting and reanalysis products
- Assess model skill against *in situ* and satellite observations to support product quality information

Research Engineer | Sorbonne University

Villefranche-sur-Mer - FR - Oct 2025 – Sep 2026

- Built and maintained data acquisition and QC pipelines for high-resolution multi-/hyper-spectral *in situ* and satellite data across two large projects
- Developed and published open-source ML algorithms to autonomously detect anomalous surface features in coastal waters via satellite imagery
- Sourced and integrated *in situ* observations to validate high-dimensional model/reanalysis data products

Co-Founder, Director | Ecomonitor

Nice - FR - Jul 2024 – Sep 2025

- Co-founded and ran an environmental data analytics startup, translating academic climate/biodiversity science into client-facing tools
- Built a library of Python/R workflows automating environmental dataset sourcing, analysis, and report generation

Lead Data Scientist | FACE-IT

Villefranche-sur-Mer - FR - Nov 2020 – Jul 2024

- Led daily operations of a multi-institution EU work package studying climate change impacts on Arctic fjord socio-ecological systems
- Sourced, harmonised, and published thousands of disparate datasets; authored the open FjordLight dataset and companion R package

ADDITIONAL

- Contributing author, WMO and NOAA annual climate reports
- Member, CLIVAR and Ilico marine heatwave research groups
- Steering committee, North-Heat project